

Last name _____

First name _____

LARSON—MATH 310—CLASSROOM WORKSHEET 03
Linear Functions.

Review

1. What is $\vec{v} + \vec{u}$?
2. What is $\vec{v} - \vec{u}$?
3. What is $4\vec{v}$?
4. What is a *linear combination*?
5. What is the definition of the *inner product* of vectors $\vec{v}$ and $\vec{u}$.
6. What is the notation for inner products?

New (from Chp.1)

1. **Transforming between two encodings for Boolean vectors.** A Boolean n -vector is one for which all entries are either 0 or 1. Such vectors are used to encode whether each of n conditions holds, with $a_i = 1$ meaning that condition i holds. Another common encoding of the same information uses the two values -1 and $+1$ for the entries. For example the Boolean vector $(0, 1, 1, 0)$ would be written using this alternative encoding as $(-1, +1, +1, -1)$. Suppose that x is a Boolean vector with entries that are 0 or 1, and y is a vector encoding the same information using the values -1 and $+1$. Express y in terms of x .
2. **(Notation)** What is $\mathbb{R}$?
3. **(Notation)** What is $\mathbb{R}^n$?
4. Consider the function $f(x, y) = 2x + 3y$. Check that f is *additive*; that is, show:
$$\text{For any pairs } (x_1, y_1) \text{ and } (x_2, y_2), f((x_1, y_1) + (x_2, y_2)) = f(x_1, y_1) + f(x_2, y_2).$$

5. Consider the function $f(x, y) = 2x + 3y$. Check that f is *homogenous*; that is, show:

$$\text{For any pair } (x, y) \text{ and scalar } \alpha, f(\alpha(x, y)) = \alpha f(x, y).$$

A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is *linear* if it is additive and homogenous. This can be shortened to:

$$\text{For any vectors } \vec{v}, \vec{w}, \text{ and scalars } \alpha, \beta, f(\alpha\vec{v} + \beta\vec{w}) = \alpha f(\vec{v}) + \beta f(\vec{w}).$$

6. Show that the function $f(x, y) = 2x + 3y$ is linear.
7. How can we write the linear function $f(x, y) = 2x + 3y$ as an inner product?
8. Why do all linear functions go through the origin? (If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a linear function, show $f(\vec{0}) = 0$).
9. Why is the function $f(x, y) = 2x + 3y + 5$ *not* linear?
10. Why is the function $f(x, y) = 2x + 3y + 5$ an *affine* function?